WORKED SOLUTION

STEP I 2019, Question 3

Pure • Integration

STEP I 2019, Question 3

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Multiplying top and bottom by $1-\sin x$ turns $1+\sin x$ into $1-\sin^2 x=\cos^2 x$, and everything over a power of $\cos x$ splits into standard pieces: $\sec^2 x$, $\sec x\tan x$, $\tan^2 x\sec^2 x$ and $\sin x/\cos^4 x$, each of which we can integrate on sight. The other two integrals are the same trick in disguise: the second becomes a $1+\cos x$ integral, which the substitution $x=\frac{1}{2}\pi-u$ turns into the first kind, and the third squares the factor $1-\sin x$.

Solution

The first integral

Since $(1+\sin x)(1-\sin x)=1-\sin^2 x=\cos^2 x$,

$$\frac{1}{1+\sin x}=\frac{1-\sin x}{\cos^2 x}=\sec^2 x-\sec x\tan x.$$

We know $\frac{\mathrm{d}}{\mathrm{d}x}\tan x=\sec^2 x$ and $\frac{\mathrm{d}}{\mathrm{d}x}\sec x=\sec x\tan x$, so

$$\int\frac{1}{1+\sin x}\,\mathrm{d}x=\tan x-\sec x+C. \qquad (*)$$

Using $\tan\frac{1}{4}\pi=1$, $\sec\frac{1}{4}\pi=\sqrt{2}$, $\tan 0=0$ and $\sec 0=1$:

$$\int_0^{\frac{1}{4}\pi}\frac{1}{1+\sin x}\,\mathrm{d}x=(1-\sqrt{2})-(0-1)=2-\sqrt{2}.$$

(Multiplying by $1-\sin x$ is fine because $\sin x<1$ throughout the range, so we never multiply by zero.)

The second integral

Multiplying top and bottom by $\cos x$ (non-zero on $[\frac{1}{4}\pi, \frac{1}{3}\pi]$),

$$\frac{1}{1+\sec x} = \frac{\cos x}{\cos x + 1} = 1 - \frac{1}{1+\cos x}. \quad (*)$$

Ask yourself How can we use $(*)$ on $\dfrac{1}{1+\cos x}$?

Since $\cos x = \sin(\frac{1}{2}\pi - x)$, substitute $u = \frac{1}{2}\pi - x$, so $\mathrm{d}x = -\mathrm{d}u$; when $x = \frac{1}{4}\pi$, $u = \frac{1}{4}\pi$, and when $x = \frac{1}{3}\pi$, $u = \frac{1}{6}\pi$. Then

$$\begin{aligned}\int_{\frac{1}{4}\pi}^{\frac{1}{3}\pi} \frac{1}{1+\cos x}\,\mathrm{d}x = \int_{\frac{1}{6}\pi}^{\frac{1}{4}\pi} \frac{1}{1+\sin u}\,\mathrm{d}u &= \Big[\tan u - \sec u\Big]_{\frac{1}{6}\pi}^{\frac{1}{4}\pi} \\ &= (1-\sqrt{2}) - \Big(\frac{1}{\sqrt{3}} - \frac{2}{\sqrt{3}}\Big) = 1 - \sqrt{2} + \frac{1}{\sqrt{3}},\end{aligned}$$

using $\tan\frac{1}{6}\pi = 1/\sqrt{3}$ and $\sec\frac{1}{6}\pi = 2/\sqrt{3}$. The interval has length $\frac{1}{3}\pi - \frac{1}{4}\pi = \frac{1}{12}\pi$, so

$$\int_{\frac{1}{4}\pi}^{\frac{1}{3}\pi} \frac{1}{1+\sec x}\,\mathrm{d}x = \frac{\pi}{12} - \Big(1 - \sqrt{2} + \frac{1}{\sqrt{3}}\Big) = \frac{\pi}{12} + \sqrt{2} - 1 - \frac{\sqrt{3}}{3}.$$

(Equally, one can multiply $\frac{1}{1+\cos x}$ by $\frac{1-\cos x}{1-\cos x}$ to get $\operatorname{cosec}^2 x - \cot x \operatorname{cosec} x$, with antiderivative $\operatorname{cosec} x - \cot x = \tan\frac{1}{2}x$; since $\tan\frac{1}{8}\pi = \sqrt{2}-1$ this gives the same answer.)

The third integral

Now we multiply top and bottom by $(1-\sin x)^2$, so the denominator becomes $\cos^4 x$:

$$\frac{1}{(1+\sin x)^2} = \frac{(1-\sin x)^2}{\cos^4 x} = \frac{1+\sin^2 x}{\cos^4 x} - \frac{2\sin x}{\cos^4 x}.$$

For the first piece, write $1 = \sin^2 x + \cos^2 x$ in the numerator:

$$\frac{1+\sin^2 x}{\cos^4 x} = \frac{\cos^2 x + 2\sin^2 x}{\cos^4 x} = \sec^2 x + 2\tan^2 x \sec^2 x.$$

Ask yourself What do $\tan^2 x \sec^2 x$ and $\sin x/\cos^4 x$ integrate to?

Both are of the form $f(x)^n f'(x)$. With $f = \tan x$, $\frac{\mathrm{d}}{\mathrm{d}x}\tan^3 x = 3\tan^2 x \sec^2 x$; with $f = \cos x$, $\frac{\mathrm{d}}{\mathrm{d}x}(\cos x)^{-3} = 3\sin x\,(\cos x)^{-4}$. Hence

$$\int \frac{1}{(1+\sin x)^2}\,\mathrm{d}x = \tan x + \tfrac{2}{3}\tan^3 x - \tfrac{2}{3}\sec^3 x + C.$$

At $x = \frac{1}{4}\pi$: $\tan = 1$ and $\sec^3 = (\sqrt{2})^3 = 2\sqrt{2}$, giving $1 + \frac{2}{3} - \frac{4}{3}\sqrt{2}$. At $x = 0$: $0 + 0 - \frac{2}{3}$. Therefore

$$\int_0^{\frac{1}{4}\pi} \frac{1}{(1+\sin x)^2}\,\mathrm{d}x = \frac{5}{3} - \frac{4\sqrt{2}}{3} + \frac{2}{3} = \frac{7-4\sqrt{2}}{3}.$$

Checks

- Size: on $[0, \frac{1}{4}\pi]$ the first integrand falls from 1 to $2-\sqrt{2} \approx 0.59$, so the integral lies between 0.46 and 0.79; $2-\sqrt{2} \approx 0.586$ fits. Likewise the third integrand is the square of the first, so its integral should be smaller: $(7-4\sqrt{2})/3 \approx 0.448$.
- Differentiate your antiderivatives: $\frac{\mathrm{d}}{\mathrm{d}x}(\tan x - \sec x) = \frac{1-\sin x}{\cos^2 x} = \frac{1}{1+\sin x}$.
- Simpson's rule gives 0.585786, 0.098663 and 0.447715 for the three integrals, matching $2-\sqrt{2}$, $\frac{\pi}{12} + \sqrt{2} - 1 - \frac{\sqrt{3}}{3}$ and $\frac{1}{3}(7-4\sqrt{2})$ to ten decimal places.

Common slips

- Sign errors in $\frac{\mathrm{d}}{\mathrm{d}x}\sec x$ or $\frac{\mathrm{d}}{\mathrm{d}x}\sec^3 x$: both carry a plus sign, as $\frac{\mathrm{d}}{\mathrm{d}x}(\cos x)^{-n} = n\sin x(\cos x)^{-n-1}$.
- Forgetting the lower limit in the first and third integrals: $\sec 0 = 1$, not 0.
- Swapping the limits wrongly after $u = \frac{1}{2}\pi - x$: the minus sign from $\mathrm{d}x = -\mathrm{d}u$ is used to put them back in increasing order.

Beyond STEP

The trick of multiplying by a "conjugate" is a special case of a systematic method: the substitution $t = \tan\frac{1}{2}x$ turns any rational function of $\sin x$ and $\cos x$ into a rational function of t, which partial fractions can then integrate. (Our answer $\tan\frac{1}{2}x$ for $\int \frac{\mathrm{d}x}{1+\cos x}$ is a hint of this.) First-year analysis also asks when an integral like these exists at all, for example near $x = -\frac{1}{2}\pi$ where $1+\sin x$ vanishes. For both ideas, see Bartle and Sherbert, *Introduction to Real Analysis*, and for many more integrals in this style, Siklos, *Advanced Problems in Mathematics*.

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